25CS2103E • DATA STRUCTURES AND ALGORITHMS (DSA-3)

Word-Level Plagiarism Checker

for Short Documents

Detecting copied and paraphrased text with hashing and dynamic programming — no plagiarism libraries used.

Team: [Student Name(s)] **Year:** 2nd Year **Guide:** [Faculty Name]

The Problem

Why copy-paste checks are no longer enough

Plagiarism in academic and professional writing undermines integrity — but naive string comparison misses reworded, reordered, or synonym-swapped copying. Commercial tools exist, but they are closed-source, cloud-dependent, and offer no insight into how similarity is actually computed.

> *“ The real gap isn’t detecting copies — it’s catching the ones that were reworded to slip past a simple text match. ”*

Exact-match tools fail

Simple diff/copy checks miss reordered or reworded passages.

Commercial tools are opaque

Cloud-based, closed-source — no visibility into the algorithm.

Costly for classroom use

Licensing and connectivity requirements limit academic use.

Our Approach

A lightweight, explainable plagiarism checker built from first principles

The system detects both verbatim copying and paraphrased or reordered passages in short documents — without relying on any prebuilt string-matching or plagiarism-detection libraries. Every matching and scoring step is hand-implemented, combining two classical algorithmic techniques:

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Rabin–Karp Hashing

Catches exact, verbatim copying

A rolling hash slides over token windows, letting the system spot copied passages in near O(n+m) time.

#

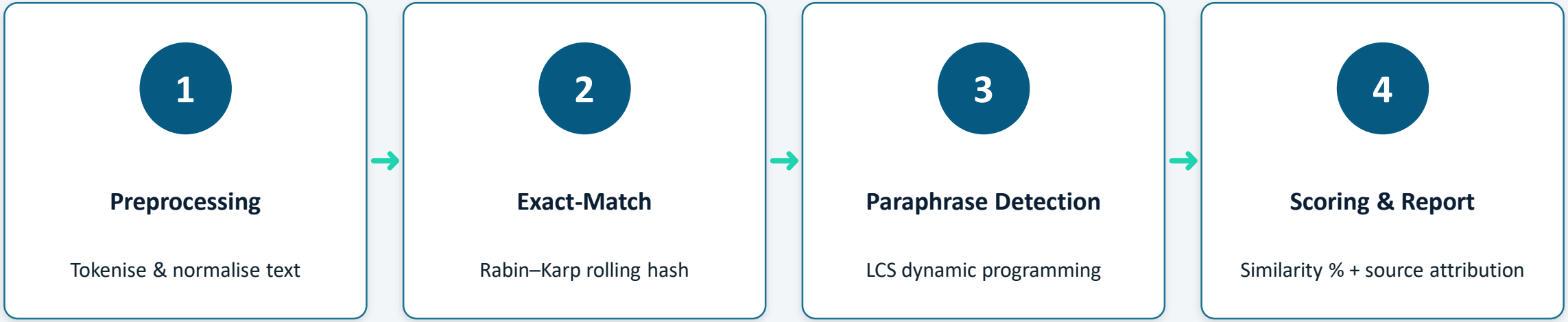
LCS Dynamic Programming

Catches paraphrased & reordered text

Sequence alignment finds the longest common subsequence between documents, surfacing reworded overlaps that hashing alone would miss.

How It Works

Four modules, run in sequence, from raw text to a similarity report



Output: each flagged span is highlighted, attributed to its likely source document, and rolled up into an overall similarity score.

Project Highlights

Built entirely from core DSA concepts — no shortcuts, no black boxes

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Core algorithms combined (Rabin–Karp + LCS)

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Prebuilt plagiarism or diff libraries used

10

Week suggested timeline for a 4-member team

3

Concepts reinforced: hashing, strings, DP

Evaluated against a hand-labelled test set of original, copied, and paraphrased documents — measuring precision, recall, and false-positive rate.

Thank You

Questions & Discussion